Invited Paper
Package Modeling and Analysis for Heterogeneous Integration

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ABSTRACT
Advanced semiconductor packaging is now used by semiconductors companies to meet both cost and performance requirements (e.g. bandwidth, latency, power, etc.) for artificial intelligence (AI) applications using heterogeneous multi-chiplet architectures. These packaging techniques include flip-chip, wafer-level packaging (both fan-in and fan-out) and 3D-Heterogeneous Integration (3D-HI) using package-on-package and hybrid/direct bonding. As detailed in several packaging roadmaps, chip-package-board co-design supported by multi-physics modeling and analysis are key underpinning technologies to support heterogeneous architectures that combine multi-chiplets within a single package.

This paper details the current trends in *Multi-physics modelling and Co-Design* for advanced semiconductor packaging. Examples detail the use of modelling techniques for (1) optimizing packaging manufacturing processes (2) electro-thermal analysis and thermo-mechanical analysis (3) predicting failures and damage in semiconductor packages when subjected to environmental stress and (4) chip-package-board co-design. The modelling techniques discussed in this paper cover both physics-based modelling and AI/Machine Learning (ML) models for design space exploration. Challenges that need to be addressed for advanced semiconductor packaging of computing applications (e.g. HPC) and developments within electronic design automation (EDA) tools and multi-physics analysis are also briefly discussed.

KEYWORDS
Advanced Semiconductor Packaging, Chiplets, Heterogeneous Integration, Co-Design, Multi-Physics Modelling, Reduced Order Modelling, and AI/Machine learning.

1 Introduction
As we enter the chiplet era, the semiconductor industry is developing advanced packaging technologies for heterogeneous integration to enable scaling and performance improvements. Hence, there is a shift from transistor scaling with the greatest number of transistors in a single large chip to package and system scaling with multiple chiplets [1-5]. But over the last 20 years, IC transistor density has increased 520x, while package I/O density has significantly lagged, increasing by only 15x [7]. To address this scaling challenge and to provide support for multi-chiplet packaged architectures, several advanced packaging and interconnection solutions have been proposed, for example, Intel (Foveros and EMIB), TSMC (Info, CoWoS) and DECA (MSeries Gen 2).

Identifying optimal packaging solutions through co-design and multi-physics modelling is critical to ensure that packaged assembly of chiplets meets system performance and reliability requirements. Chips are vulnerable to both global and local sources of thermo-mechanical stress, where global stresses result from coefficient of thermal expansion (CTE) mismatch between the chip, package, and board materials, and local stresses arise from temperature gradients due to self-heating effects within the chip. At the chip level, these can result in die cracking, under bump metallurgy (UBM) cracking, back-end-of-line (BEOL) failures, through silicon via (TSV) stress, metal migration, Cu/ELK cracking, and mobility shifts in transistor performance. At the package/board level, this can result in bump failures, redistribution layer (RDL) failures, underfill cracking, etc. Hence thermal and thermo-mechanical analysis is required across the chip-package-board domains.

2 Why Chiplets?
Gordon Moore's seminal paper – *Cramming More Components Onto Integrated Circuits* – published in 1965 [7] foresaw the limitations of transistor scaling and cramming as many of these onto a single chip. In this paper he stated "it may prove to be more economical to build large systems out of smaller functions, which are separately packaged and interconnected". Hence, in Moore's paper this statement has now become very important and essentially predicted the need for heterogeneous integration and advanced packaging, which is the cornerstone of today's chiplet technology. Traditional packaging – using wirebonds and solder interconnects to connect the semiconductor IC to the outside world – has served the community well with its primary purpose of protecting the semiconductor devices from the external environment and providing connections to other devices/packages across a printed circuit board.

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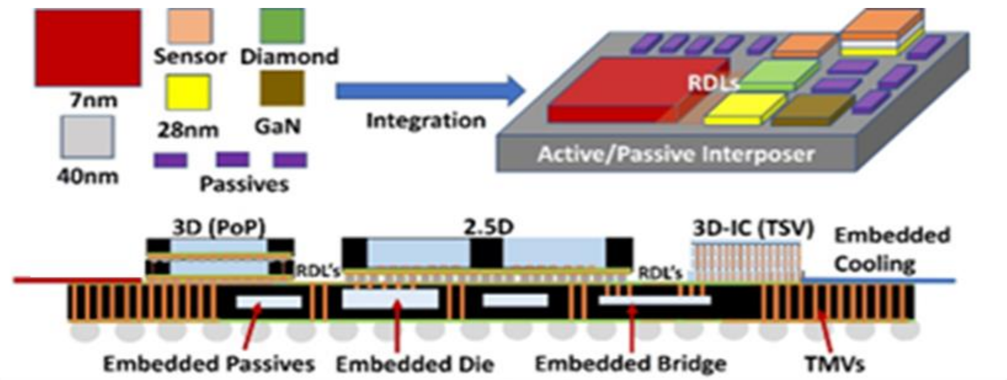

Device Packaging	Bonding	Interposers	Interconnects
Wafer Level Packaging	Flip Chip	Organics	C4, Microbump's, Cu-Pillar
3D PoP, 3D-IC (Die Stack)	Thermo-Compression	Silicon	RDL's, TxVs, Optical
SiP, QFN, BGA, Hybrids...	Hybrid Direct Bonding	Ceramics	Cu-Cu (bumpless)

Figure 1. Packaging Toolbox

Chiplet technology supported by advanced packaging provides a route for disaggregating large system on chip (SoC) designs into smaller specialized chiplets each with a specific function. Specific advantages of chiplets compared to large SoC's are:

- Chiplets can be produced at an appropriate technology node and sourced from different vendors. This provides flexibility and potentially optimized performance and cost.
- A chiplet ecosystem and market can provide much faster time to market compared to SoC.
- Yield can be significantly improved compared to monolithic 2D systems.

Although chiplets provide the above advantages, they also face challenges such as complex interconnect design within a package, thermal management issues, especially for staked die within a package, and the additional costs for advanced packaging. In terms of design flows, chiplets and heterogeneous integration can be disruptive to traditional design methods and requires new enabling technologies such as:

- Die-to-die interface IP, which are the physical files, are required to implement the high-speed interfaces within a package. Examples include UCIe, bunch of wires (BoW), and advanced interface bus (AIB), which contain protocols for low power to drive these signals.
- Innovations in packaging technology are required to address the need for effective cooling as well as the architecture to support high bandwidth and low latency performance.
- EDA tool development is required to support design workflows for these 2.5D and/or 3D structures and provide close linkages between chip-package-board designs (architecture planning/analysis, partitioning, floorplanning, routing, power delivery, timing analysis, etc.) as well as signal integrity/power integrity analysis (SI/PI), thermal, and mechanical analysis; and design for test infrastructure.

3 Advanced Semiconductor Packaging

As detailed in Section 2, chiplets are the building blocks to a system that can be assembled to provide an integrated system of data storage, signal processing, and computing with other IP components connected through a packaging architecture. In a chiplet system, it is possible to have a processing unit, an AI accelerator, and stacks of memory all communicate and share data as if they were all on the same chip, but by moving memory closer to the processor, it can help tackle bigger tasks (for example, AI computing) whilst saving more energy.

As illustrated in Figure 1, there are many ways you can integrate chiplets into a package. For example, stacking them onto each other through vertical interconnections or placing them side-by-side on a substrate or interposer where they are supported through dense electrical interconnections for communication.

Chiplet architectures only work if the interconnects between them are fast enough for performance requirements. For 2.5D architectures, this involved the use of silicon interposers which are manufactured using semiconductor device manufacturing processes. Silicon interposers are expensive and limited to the reticle size. To create larger interposers, reticle stitching is required. An alternative approach is to use high-density organic substrates which are essentially high-quality PCB's that can potentially provide better thermal and mechanical properties.

To avoid the high cost and size challenges of silicon interposers, a hybrid approach can be pursued using smaller silicon bridges, which are silicon dies embedded into an organic substrate. This approach was pioneered by Intel through their Embedded Multi-Die Interconnect Bridge (EMIB) which allows high bandwidth chiplet interconnections and high-density interconnections through very small bump pitches (10um – 20um).

3D stacking of chiplets can be achieved through bonding of die using microbumps or copper-copper hybrid bonding where dies are connected vertically and through-silicon vias act as regular metal interconnects allowing for finer pitch (<10um). Examples of this approach using hybrid bonding include Intel's Foveros and TSMC's SoIC technologies.

For AI and high-performance computing (HPC) applications, a combination of 2.5D and 3D technologies for cost effective high bandwidth and low latency performance is attractive. For such architectures, fan-out-wafer-level packaging (FOWLP) offers an ideal platform for 3D-HI. In this approach, known-good-dies from different wafers can be processed and reconstituted on a carrier wafer where RDL are built up to provide very fine line/spacing and high-density interconnection between dies which are then over-moulded and solder balls are then placed onto of the RDL. This eliminates the need for a package substrate as this architecture acts as both a package and substrate which can be connected directly onto a PCB.

An example of FOWLP is the DECA M-Series Gen2 platform with adaptive patterning for routing between dies, through automated optical inspection and real-time integration with EDA tools for pathfinding and design rule checks [3]. This technology can achieve 20 µm bond-pad pitch, 2 µm L/S, and 2518 I/O/mm2, compared to Intel's EMIB (490 I/O/mm2) and TSMC's INFO packaging platforms (545 I/O/mm2) [8].

4 Multi-Physics Modelling

Modelling and simulation have seen tremendous advances over the last twenty years supported by advanced mathematical approaches (e.g. finite element analysis (FEA), multi-physics modelling, numerical optimization, AI/ML, etc.) as well as advances in computational hardware. It is now possible to develop models and undertake simulations of electronic devices, components and systems that would have been impossible five or ten years ago.

High-fidelity (e.g., FEA) models have been used extensively for predicting package performance in terms of signal/power integrity, thermal, and thermo-mechanical behavior using tools such as Ansys, Flotherm, Abaqus, etc. Optimizing package design in terms of system performance (electrical and thermal) and reliability (thermal and mechanical) requires an electrical-thermal-mechanical co-design approaches to support Design for X (X: Manufacture, Reliability, Test, etc.). Together with a multi-objective capability, these tools provide a route to explore the chip-package-board design space and to ensure that performance metrics are met, and the risk of failure modes are mitigated.

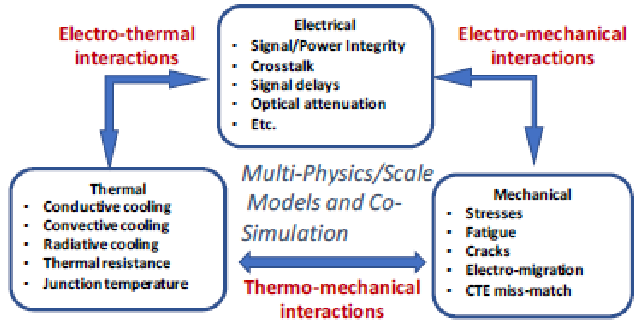

Figure 2. Interaction between physical domains

Figure 2 details the key physical phenomena that need to be predicted in the electrical, thermal and mechanical domains. Interactions at the electro-thermal interface include mapping power distribution accurately from a chip model into a temperature map in a package and system model. For the thermo-mechanical interface, examples include accurately leveraging the thermal stress resulting from a package/system model into the backend of the chip to predict physical stress and damage in the regions of the through-silicon vias (TSVs). These stresses will hence impact the layout of the TSVs as well as the transistor threshold voltages and drive currents. A classic example of electro-mechanical modelling is electro- and stress-migration in interconnects both on and off the chip. Again, package stresses need to be considered to accurately model the impact of metal migration at the chip level. These are only a small list some of the examples that demonstrate the need for co-design and multi-physics analysis across the chip-package-system domains, and the need to accurately model the interactions between electrical, thermal and mechanical domains.

4.1 Manufacturing Process Modelling

Packaging processes such as building RDL, molding, and bonding can induce stresses and failures that will impact overall quality of the packaged assembly and hence subsequent reliability. These processes are highly non-linear where high-fidelity simulations require accurate process dependent material properties such as viscoelastic flow properties for simulating a molding process. Hence the strong link between metrology and modelling is required to ensure accurate model predictions.

An example of process modelling of packaging processes includes predictions of warpage that occurs during the fan-out-wafer-level packaging (FOWLP) process steps as shown in Figure 3. Ensuring that warpage is controlled is a major requirement for FOWLP as the consequences for high warpage impacts subsequent process conditions and overall reliability of the package due to chip-cracking, bonds detaching, and RDL delaminating.

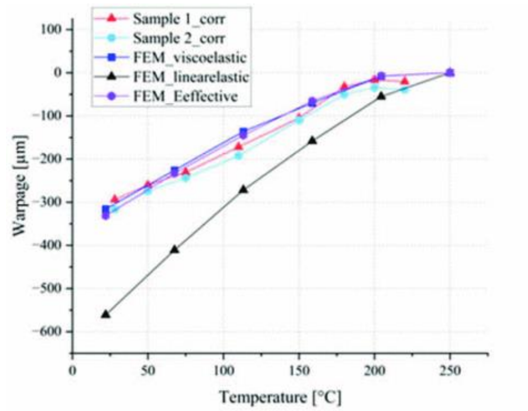

Figure 3. Modelling Warpage in FOWLP process: comparison between linear and non-linear models [8]

Numerical models for the FOWLP process require high precision simulations to predict the influence of different materials and process parameters on warpage. Such precision requires non-linear material behavior (e.g. viscoelastic behavior of polymer materials). Using this approach, simulations have investigated the sensitivity of the polyimide material properties and copper layer designs on warpage and showed that assuming elastic behavior of the polyimide can lead to modelling errors of >80% and hence non-linear viscoelastic properties are required [9].

Another key process that can impact package quality and hence reliability are interconnect bonding process such as flip-chip bonding for die-to-die stacking. A key requirement here is to understand the impact of alignment accuracy on the quality of the fabricated bonds for high interconnect applications. Techniques such as solder micro-bumps formed during a reflow process, thermo-compression bonding, and copper-copper hybrid bonding can be used depending on the bump pitch required.

An example of numerical modelling of the flip-chip bonding process has investigated both reflow soldering and thermo-compression bonding of indium bumps for focal plane arrays where a compound semiconductor pixel array IC is bonded to a silicon read-out IC die [10]. As detailed in Figure 4, the impact of misalignment can have serious consequences on bond quality where larger misalignment leads to bump collapse and very small stand-off heights for the formed bonds. This will lead to very poor subsequent reliability due to CTE mismatch in the materials. This work demonstrated that for ultra-fine pitch applications an alignment accuracy of sub-um was required to avoid bump collapse.

The above models also demonstrated the feasibility of using reflow soldering as the assembly process where the surface tension restoring force can be used to address misalignment. Although feasible for high bump pitches, the work demonstrated that for ultra-fine pitches the CTE mismatch between the dies and the subsequent relative displacement between them will result in solder bridging where two interconnects collide.

For advanced packaging of multi-chiplet architectures, there is a need for further modelling of key manufacturing processes. These are highly non-linear models where accurate materials data is required to capture material behavior accurately during each process step.

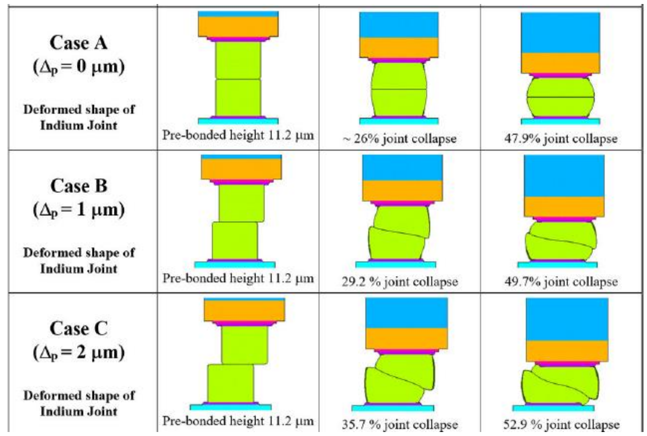

Figure 4. Modelling thermo-compression bonding process for die-die stacking focal plan array application [10]

4.2 Themal/Electro-Thermal Analysis

Placing more chiplets into a smaller volume means that power density will increase, also contributing to more hotspots with higher temperature. New solutions are required for thermal management of advanced packages combining both passive and active cooling. Careful consideration on chip-level is also required in terms of hotspot management and ensuring that cooling solutions address this. Polymer based thermal interface materials (TIM), which have relatively low thermal conductivity, are still the main medium for moving heat across packaging interfaces to a heat spreader, although in future, metallic TIMs may be required. For the design-driven solutions in 3D-HI, numerous embedded cooling structures are compared, including thermal via [11], heat pipe [12], and microchannel [13].

To effectively optimize the thermal solution, there is a need to predict hotspots and model the heat transfer from proposed design solutions to an external heat spreader. This requires electro-thermal modelling across the chip-package domain. For reduced order thermal models, an example include liquid-cooling microchannel design analysis and optimization [14,15].

However, as the need for system-level analysis grows, EDA transitions from incorporating single physics to multi-physics. Examples include rapid mapping of temperature to IR drop distribution enabling multi-physics power grid analysis [16]. Commercial tools are also available from Ansys (Redhawk, Fluent), Cadence, Siemens, COMSOL, etc., which yet covers up to two physics at most. For complete analysis incorporating three or more physics phenomena as shown in Figure 2, with uncertainty consideration caused by chip-package-system process variations, an integrated flow of multiple packages will be required. Along with advanced numerical approach, ML methodologies are simultaneously being intensively researched, further discussed in Section 5.

4.3 Thermo-Mechanical Analysis

Thermally induced stresses occur both in the operation of a semiconductor package and during its testing. This is the most significant reason for early failures of semiconductor packaging due to interconnect fatigue. Hence as detailed above, effective thermal management is critically important to ensure the expected reliability of the package. There has been extensive modelling of solder joint interconnects in semiconductor packaging both within the package (component level) and between the package and the PCB.

After a package is fabricated, it will undergo component level and board level qualification as per the JEDEC specifications. These tests expose the package and board to thermal, vibration, and humidity induced stresses. Using modelling to virtually assess a package's thermo-mechanical performance before physical prototyping can result in extensive cost savings as well as ensure a package design and architecture meets the required reliability specifications. Figure 5 details an example of a solder bump model at the package-PCB interface. In this analysis, the design of the PCB and the use of additional material such as underfills and edgebonds were investigated to minimize the risk of early fatigue damage to these interconnect structures. Simulation results show that the highest plastic work (an indicator for crack initiation and fatigue failure) occurs at the solder bump/PCB interface for this design.

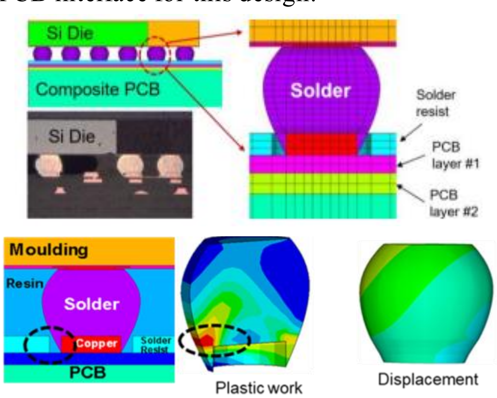

Figure 5. Solder interconnect models and predictions of plastic work when subjected to a thermal load [17]

Figure 6 details model predictions for different PCB's and clearly identifies the impact of both edgebonds and underfills as options to stress relieve the solder joints when subjected to thermal loads and hence minimizes the risk of early fatigue failure.

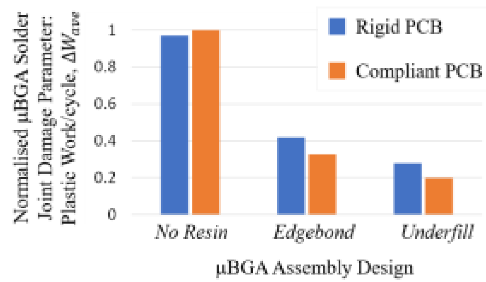

Figure 6. Model predictions of damage in the solder interconnects and impact of different material choices [17]

Table 1. Reliability test data for the solder joints modelled above, where N63.2% is a measure of characteristic life of the package samples being tested [17].

Micro-BGA Assembly			**Reliability Test Data**		
Ref	***PCB Type***	***Resin***	***95% Confidence Low Limit***	$N_{63.2\%}$	***95% Confidence Upper Limit***
3	Rigid	No resin	1,142	1,220	1,303
6	Rigid	Edgebond	3,592	3,991	4,433
8	Rigid	Underfill	8,255 cycles with no failure		
10	Compliant	No resin	917	948	980
14	Compliant	Edgebond	7,926 [a]	8,855 [a]	9,894 [a]
18	Compliant	Underfill	6,923 cycles with no failure		

Table 1 details subsequent reliability test data showing the trends predicted by the numerical models above being followed and the significant benefit of using an underfill material between the package and PCB.

Modelling the thermo-mechanical behavior of very small interconnects requires a multi-scale modelling approach where the impacts of grain structure in the material can govern its overall behavior. This requires coupling micro models of grain structure with macro-models of the interconnect [18]. Other physics that generates stress can also be modelled using finite element techniques such as vibration [19]. Challenges that need to be addressed for advanced semiconductor packaging architectures is the development of accurate lifetime models for failure modes such as electro-migration which again require a multi-physics/multiscale computational capability.

5 Need for AI/ML and Physics-informed AI/ML

While FEA provides high-fidelity data, the computational cost exponentially increases with design complexity and proportionally increases with the number of designs to compare. Alternative computation approach is ML-driven methodologies, which are renowned for the ability to model complex physical phenomena. Moreover, it has exceptional inference ability to predict the performance of unknown designs based on trained model of similar designs, enabling meaningful design space exploration with reduced computational time and cost. Existing work includes predictive ML model for novel material (e.g. Cu-CNT composites) electrical and thermal properties [20], reduced order thermal correlations of heat sinks [21,22].

As the complexity of package design increases and multiple physics are involved in a system, computational time of numerical simulations becomes the bottleneck in obtaining sufficient ML training dataset. With limited dataset, traditional data-driven ML approaches can lead to overfitting models and can provide infeasible results, which disagree with physics laws. To leverage the limitations, physics-informed neural network (PINN) [23], illustrated in Figure 7, is rapidly gaining attention, that embeds scientific governing equations into neural network to reduce the solution space to feasible range, leading to more accurate models. PINN has demonstrated promising results on predicting heat transfer performance (forward problem) and reconstructing temperature, pressure, fluid velocity distributions using sparse data (inverse problem) [24].

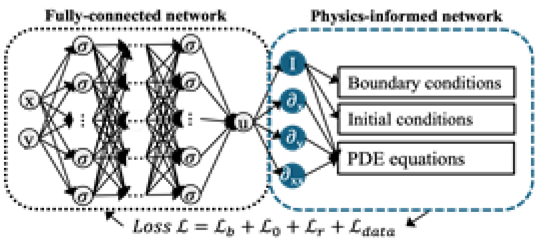

Figure 7. PINN architecture

Despite the breakthrough of PINN, there yet remains challenges which includes building generalized, scalable model with reliable inference capability, training on design A and predicting on design B. For example, conventional PINN is exhibiting limitations in geometry generalization, and newer architectures to capture spatial information are being researched. In addition, model robustness of training with noisy experimental data (e.g. metrology measurements) and uncertainty from process variations are not widely explored. These are crucial steps to accomplish prior to real-time analysis and optimization, that can be integrated with DECA's adaptive patterning (AP Live) and perform effective design space exploration.

6 Chip-Package-Board Co-Design

As detailed in Figure 8, traditional design tools and workflows for chip, package and system (PCB) generally used different tools and essentially followed a sequential process: chip-first, then package, then board. This leads to examples of a PCB team receiving chip and package designs that are PCB-unfriendly in terms of pin assignments leading to issues with routing that cannot meet requirements. All the major EDA companies are working on new developments for chip-package-board co-design and the trend is for these design flows to be physics informed.

Given accurate multi-physics modeling of the package system is available (e.g. thermomechanical performance, warpage, signal integrity, electromigration), these results can further be feedback to physical chip design for co-design optimization. One of the early works for advanced package-driven co-design is warpage-aware floorplanning [25].

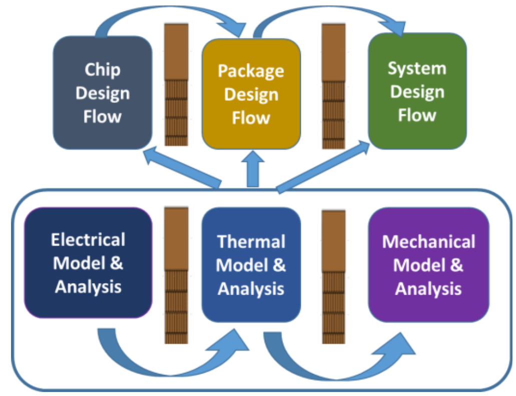

Figure 8. Traditional disjointed approach to design and analysis

8 Conclusions

Ever since the 1960's when the first spice tools became available and the growth in the use of FEA started, there have been developments in modelling and simulation tools for a wide range of applications. Traditionally these tools were focused on distinct physics such as electromagnetics, thermal, or mechanical stress analysis. Today, we refer to the simulation and modelling tools as multi-physics and multi-scale and use terms such as co-design and co-simulation. As detailed above, co-design, multi-physics modelling and simulation are underpinning technologies for future heterogeneous integrated electronic systems. Example challenges that need to be addressed are:

- **Multi-physics/scale models**: Interactions across the chip-package-system domains need to be modelled accurately. These will need to address range of geometric features at the nm-scale (chip) to the cm-scale (package, system).
- **Fast Solvers**: What level of model abstraction should be used at the chip, package, and system domains? There is a need for reduced order modelling and AI/ML techniques that capture nonlinearity and predict key physics in real time.
- **Multi-objective/variate Optimization under Uncertainty**: Complexity multi-physics domain interactions require robust multi-objective optimization solvers that can handle large number of variables and design constraints and provide solutions in a computationally fast manner.
- **Lifetime Models**: Heterogeneous integrated systems will require new physics-of-failure models.
- **Physics informed co-design:** Integrated chip-package-board co-design that is physics informed. For example, thermo-mechanically informed floorplanning and routing.

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